Blessed G/Michael Catholic Secondary School

G-10 Physics Test 2 **Date: 07/04/2018 EC** **Time allowed: 1:00 hr**

Name: Bitanya Wondimagegnehu **Grade &sec:** 10th B **Number:** 12

I. Match the column A with column B correctly.

A.	B.
C 1. The slope of position – time graph.	A. displacement
D 2. The slope of velocity – time graph.	B. velocity
A 3. The area under velocity – time graph.	C. change in velocity
B 4. The area under acceleration – time graph.	D. acceleration
	E. change in acceleration

II. For each of the following questions, choose the best answer from the given alternatives.

C 5. If a ball is dropped from rest, it will fall 20 m during the first two seconds. How far will it fall during the third and fourth seconds?

A. 20 m B. 40 m C. 60 m D. 80 m

A 6. If a car requires 5 seconds to accelerate from zero to 90 km per hour, what is its average acceleration?

A. 18 m/sec^2 B. 9 m/sec^2 C. 10 m/sec^2 D. 5 m/sec^2

C 7. The average speed of an object is defined to be

A. distance it travels multiplied by the time it takes.

B. the distance it travels divided by the time it takes.

C. one half of the sum of the maximum and the minimum speeds.

D. the speed determined over an infinitesimally small time interval.

D 8. A car initially traveling north at 5 m/s has a constant acceleration and reaches a velocity of 20m/s north. During this time, the car has traveled a displacement of 50 m north. How long does it take to travel this displacement?

A. 10 s B. 8 s C. 5 s D. 4 s

A 9. In straight line motion the

A. acceleration is parallel (or antiparallel) to the velocity.

B. acceleration is perpendicular to the velocity.

C. acceleration is vertical, while the velocity can be in any direction.

D. acceleration is vertical and the velocity is horizontal.

C 10. A body moves from rest with a uniform acceleration and travels 27 m in 3 s. What is the acceleration of the body?

A. 18 m/sec^2 B. 9 m/sec^2 C. 6 m/sec^2 D. 3 m/sec^2

AC 11. A body moves from rest with a uniform acceleration and travels 27 m in 3 s. What is the velocity of the body at 4 s after the start?

A. 72 m/s B. 36 m/s C. 24 m/s D. 9 m/s

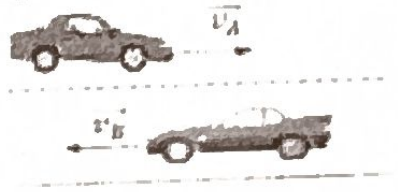
A 12. A train moving with a velocity of 20 m/s is brought to rest by applying brakes in 5 s. How far it travelled during the retardation?

A. 50 m B. 100 m C. 4 m D. 8 m

B 13. A ball dropped from a height of 45 m building. Assuming the acceleration due to gravity is g =10 m/s^2 and neglecting air resistance, how long will it take to reach the ground?

A. 4.5 sec B. 3 sec C. 6 sec D. 49 sec

C 14. Two cars, A and B are traveling with the same speed of 90 km/h in opposite directions, as shown in the figure. What is the relative velocity of car A with respect to car B?



A. 0 km/h C. 180 km/h, right

B. 180 km/h, left D. 45 km/h, right E. 45 km/h, left

B A 15. Average speed of a car between points A and B is 20 m/s, between B and C is 15 m/s, between C and D is 10 m/s. What is the average speed between A and D, if the time taken in the mentioned sections is 5s, 10s and 15s respectively?

A. 15.0 m/s B. 13.3 m/s C. 17.3 m/s D. 25.0 m/s

Use the velocity-time graph to the right to answer questions 16 through 21.

D C 16. At what time did the object change direction?

A) 0 s B) 10s C) 35 s D) 55 s

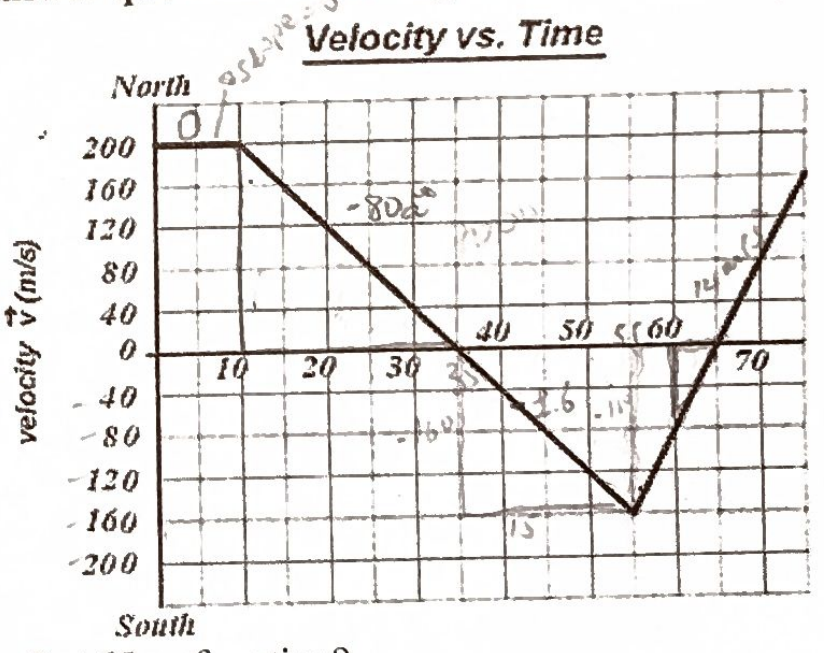
C 17. During what time interval did the object speed up?

A) 0 s to 10 s C) 35 s to 55 s

B) 10 s to 35 s D) 55 s to 65 s

C A 18. At what time was the object's acceleration be zero?

A) 5 s B) 15 s C) 35 s D) 55 s



B D 19. How much distance does the object travel in the first 65 s of motion?

A) 2100 m B) 2500 m C) 4500 m D) 6900 m

A 20. What is the magnitude of the displacement does the object travel in the first 65 s of motion?

A) 2100 m B) 2500 m C) 4500 m D) 6900 m

C 21. During what time interval did the object slow down?

A) 0 s to 10 s B) 10 s to 55 s C) 55 s to 65 s D) 65 s to 75 s